

JAINDU SAMARANAYAKE

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EDUCATION

Bachelor of Science in Electronic and Telecommunication Engineering

University of Moratuwa, Sri Lanka

March 2024 - Present

CGPA: 3.65/4.00

TECHNICAL SKILLS & INTERESTS

Hardware & Embedded

ESP32, Raspberry Pi, Arduino Mega, DE0-Nano (FPGA), I2C, UART

Engineering Software

Altium Designer, SolidWorks, Intel FPGA (Quartus), MATLAB/Simulink

Programming

C++, Python, Java

Domains of Interest

Embedded Systems, Firmware Development, IoT, Hardware Design, Robotics, ML

PROJECTS

Dengue Patient Monitoring System (Team NEXORA) | SLIoT & SPARK Challenge

November 2025 - Present

Present

- Developed a wrist-worn health monitoring node to continuously measure SpO2, blood pressure, and pulse pressure (SBP-DBP difference).
- Programmed firmware on an ESP32 microchip and established I2C sensor connections and serial communication protocols with a Raspberry Pi for central processing.
- Trained and integrated a machine learning model to analyze patient vitals locally on the hardware.

Smart Water Management System | Academic Group Project

- Designed a custom PCB buck-boost converter and LDO circuit using Altium Designer for the power supply of a digital water flow tracking device.
- Integrated a pulse output meter and a GSM module to transmit hardware telemetry data to a cloud server.

Hardware Product Development (Sound Regulation IoT) | Curriculum Project

February 2025 - November 2025

2025

- Designed an analog circuit to divide sound inputs into specific frequency bands and regulate microphone feed.
- Completed the full hardware lifecycle, including custom PCB layout and physical enclosure design.

FPGA UART Communication Protocol | Academic Project

- Implemented and tested a UART serial communication protocol utilizing a DE0-Nano FPGA development board.

Autonomous Robotics (SLRC 2026) | Competition Project

July 2025 - Present

- Developed the virtual world integration by programming a system to read AprilTags and extract spatial coordinates.
- Transmitted the telemetry data via a REST API to bridge the virtual simulation with the robot's automation logic.

Laboratory Database Management System | Academic Group Project

- Developed a backend REST API using Java Spring Boot to manage database queries and laboratory assets.
- Integrated an administrative frontend dashboard to interface with the backend system.

POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR

• **Committee Member, IEEE Robotics and Automation Society (RAS)**

November 2025 - Present

Organized MAZEX 1.0, a prominent Micromouse competition.

• **Organizing Committee Member:** ENTC Career Fair '26 (Jan 2026) & SLRC (Apr 2025)

• **Mentor, Soyuru Sathkara Seminar Series**

January 2025

REFERENCES

Dr. Ranga Rodrigo – Senior Lecturer, Dept. of Electronic & Telecommunication Engineering, University of Moratuwa
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